

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: ONE

CLASS: SS1

SUBJECT: Chemistry

TOPIC: Introduction to Chemistry

Introduction

- Chemistry is the study of matter — its composition, structure, properties and the changes it undergoes. It is called the central science because it links physics, biology and every applied science.

Why chemistry matters

- It explains everyday life: cooking, rusting, burning, medicine, soap, fertiliser, food preservation and the treatment of water. Chemistry has produced drugs that cure diseases, improved farming yields and provided materials like plastics, glass and cement.

Branches of chemistry

- Physical chemistry — the energy, rates and states of matter (kinetics, thermodynamics).
- Organic chemistry — carbon compounds (fuels, plastics, drugs, food).
- Inorganic chemistry — metals, non-metals, acids, bases and salts.
- Analytical chemistry — identifying what substances are present and how much (testing water purity).
- Biochemistry — chemical processes in living things (enzymes, DNA, metabolism).

CLASS EVALUATION

1. Define chemistry and state the main branches.
2. Mention three ways chemistry improves everyday life.
3. Why is chemistry called the central science?

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: TWO

CLASS: SS1

SUBJECT: Chemistry

TOPIC: Laboratory Safety & Apparatus

Introduction

- A laboratory is a room set aside for practical chemistry. Safety rules and correct use of apparatus protect the learner and make experiments reliable.

Safety rules

- Always wear a lab coat and goggles; never taste or smell chemicals directly (waft).
- Handle acids and alkalis carefully; wash splashes with plenty of water.
- Never point a heated test tube at anyone; heat test tubes at an angle. Keep the bench tidy; tie back hair; report every accident at once.

Common apparatus and uses

- **Beaker** — holding and mixing liquids; **conical flask** — mixing and titration.
- **Measuring cylinder** — accurate volume of liquids; **burette and pipette** — precise volumes in titration.
- **Test tube and boiling tube** — small-scale reactions and heating; **funnel** — filtering and pouring.
- **Evaporating dish** — evaporating solutions to dryness; **crucible** — strong heating of solids.
- **Bunsen burner** — heating; **tripod stand, gauze and retort stand** — supporting apparatus.
- **Watch glass, spatula, glass rod, wash bottle** — covers, sampling and stirring.

CLASS EVALUATION

1. List five safety rules in the laboratory.
2. Which apparatus would you use to measure 25 cm³ of water accurately?
3. State the use of a crucible and an evaporating dish.
4. Why must chemicals never be tasted?

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: THREE

CLASS: SS1

SUBJECT: Chemistry

TOPIC: Matter & Separation

Introduction

- Matter is anything that has mass and occupies space. It exists in three physical states — solid, liquid and gas — which can change into one another by heating or cooling.

States of matter

- **Solid** — fixed shape and volume; particles closely packed, vibrating about fixed positions.
- **Liquid** — fixed volume but takes the shape of its container; particles close but able to slide.
- **Gas** — no fixed shape or volume; particles far apart and moving rapidly and randomly.
Changes: melting, freezing, evaporation, condensation, sublimation.

Separation techniques

- **Filtration** — separates an insoluble solid from a liquid (sand from water).
- **Evaporation/crystallisation** — recovers a dissolved solid from a solution (salt from sea water).
- **Simple distillation** — separates a liquid from a solution (pure water from salt solution).
- **Fractional distillation** — separates miscible liquids of different boiling points (crude oil; ethanol–water).
- **Chromatography** — separates dyes in inks.
- **Decantation, centrifugation, sublimation and magnetic separation** — for other mixtures.

CLASS EVALUATION

1. Define matter and name its three states with particle arrangements.
2. Which technique separates a mixture of sand and salt?
3. How would you separate coloured dyes in a pen?
4. State the difference between a pure substance and a mixture.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FOUR

CLASS: SS1

SUBJECT: Chemistry

TOPIC: Atomic Structure

Introduction

- An atom is the smallest particle of an element that can take part in a chemical reaction. It consists of a nucleus surrounded by shells of electrons.

Sub-atomic particles

- **Protons** — positive, in the nucleus; their number defines the element (atomic number).
- **Neutrons** — neutral, in the nucleus; their number adds mass without changing the element.
- **Electrons** — negative, arranged in shells outside the nucleus; in a neutral atom the number of electrons equals the number of protons.

Key definitions

- Atomic number (Z) = number of protons; mass number (A) = protons + neutrons. Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons (e.g. Cl-35 and Cl-37). Electron configuration shows the arrangement of electrons in shells (e.g. sodium 2,8,1), which explains the chemical behaviour of the element.

CLASS EVALUATION

1. State the charge and location of each sub-atomic particle.
2. Calculate the number of protons, neutrons and electrons in ^{23}Na ■.
3. Define isotopes with an example.
4. Write the electron configuration of oxygen ($Z = 8$) and predict its group.

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: FIVE

CLASS: SS1

SUBJECT: Chemistry

TOPIC: Elements, Compounds & Mixtures

Introduction

- Matter is classified into elements, compounds and mixtures. An element is a substance that cannot be split into simpler substances by chemical means; a compound is formed when two or more elements combine chemically in fixed proportions.

Distinguishing features

- Elements have one type of atom (e.g. oxygen, iron, gold). Compounds have properties completely different from their elements (e.g. hydrogen + oxygen → water) and can only be separated by chemical means. Mixtures are formed by physical combination (air, salt + sand) with each component keeping its properties, and are separated by physical methods.

Common examples

- Elements: metals (iron, copper, sodium), non-metals (oxygen, carbon, sulphur).
Compounds: water (H_2O), carbon dioxide (CO_2), common salt (NaCl), lime (CaO).
Mixtures: air, sea water, brass, petroleum.

CLASS EVALUATION

1. Distinguish between an element and a compound with examples.
2. Is air a compound or a mixture? Give reasons.
3. Name two methods of separating mixtures.
4. Why do compounds differ in properties from their constituent elements?

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TERM: FIRST 2026/2027 ACADEMIC SESSION

WEEK: SIX

CLASS: SS1

SUBJECT: Chemistry

TOPIC: Separation Challenges & Laboratory Skills

Introduction

- This week consolidates the practical skill of choosing the right separation technique and recording observations accurately in a laboratory report.

Choosing a technique

- Ask: is the mixture solid + solid, solid + liquid, or liquid + liquid? Are the components soluble? Do the boiling points differ? The answers decide whether to filter, evaporate, distil or use a separating funnel (oil and water).

Laboratory report format

- **Title, aim, apparatus, method, observation, conclusion.** Observations must be factual (colour change, gas evolved, precipitate formed), never interpretations. Accuracy: read the meniscus at eye level, rinse apparatus, record results to the correct number of decimal places.

CLASS EVALUATION

1. Describe how you would obtain pure water from a salt solution.
2. Why is a separating funnel suitable for oil and water?
3. What is the difference between an observation and a conclusion?
4. State three good laboratory habits that improve accuracy.